

Unit 1: Numbers, Quantification and Numerical Applications

Modulo Arithmetic

- **Definition:** $a \equiv b \pmod{n}$ if $a - b$ is divisible by n .
- **Properties:**
 - $(a + b) \pmod{n} = ((a \pmod{n}) + (b \pmod{n})) \pmod{n}$
 - $(a \times b) \pmod{n} = ((a \pmod{n}) \times (b \pmod{n})) \pmod{n}$

Numerical Inequalities

- **Intervals:**
 - Open: $(a, b) = x \mid a < x < b$
 - Closed: $[a, b] = x \mid a \leq x \leq b$
 - Half-open: $[a, b) = x \mid a \leq x < b, (a, b] = x \mid a < x \leq b$
- **Absolute Value:** $|x| < a \implies -a < x < a; |x| > a \implies x < -a \text{ or } x > a$

Alligation and Mixture

- **Rule of Alligation:** Ratio of quantities of two ingredients mixed = $\frac{\text{Cost of cheaper} - \text{Mean Cost}}{\text{Mean Cost} - \text{Cost of dearer}}$

Boats and Streams

- **Downstream Speed:** $V_d = u + v$ (where u = speed of boat in still water, v = speed of stream)
- **Upstream Speed:** $V_u = u - v$
- **Speed of boat in still water:** $u = \frac{1}{2}(V_d + V_u)$
- **Speed of stream:** $v = \frac{1}{2}(V_d - V_u)$

Pipes and Cisterns

- If a pipe fills a tank in x hours, it fills $\frac{1}{x}$ of the tank in 1 hour.
- If a pipe empties a tank in y hours, it empties $\frac{1}{y}$ of the tank in 1 hour.
- If both work together: $\frac{1}{x} - \frac{1}{y}$ (for filling and emptying)

Races and Games

- **A gives B a start of x meters:** A starts x meters behind B.
- **A can give B x points:** A scores x points more than B to win.

Unit 2: Algebra

Matrices

- **Order:** $m \times n$ (rows x columns)
- **Types:** Row, Column, Square, Diagonal, Scalar, Identity (I), Zero (O)
- **Operations:**
 - **Addition/Subtraction:** Element-wise (same order)
 - **Scalar Multiplication:** $kA = [ka_{ij}]$
 - **Matrix Multiplication:** $\$(AB)\{ij\} = \sum_k A\{ik\} B_{kj}\$$ (Number of columns in A = Number of rows in B)
 - **Transpose:** $\$(A^T)\{ij\} = A\{ji\}; (AB)^T = B^T A^T \$$
- **Determinant:**
 - For 2×2 : $\det(A) = ad - bc$ for $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$
 - For 3×3 : Use cofactor expansion.
- **Inverse:** $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$ (exists if $\det(A) \neq 0$)
 - For 2×2 : $A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

Relations and Functions

- **Relations:** Set of ordered pairs.
- **Types of Relations:**
 - **Reflexive:** $(a, a) \in R$
 - **Symmetric:** $(a, b) \in R \implies (b, a) \in R$
 - **Transitive:** $(a, b) \in R$ and $(b, c) \in R \implies (a, c) \in R$
 - **Equivalence Relation:** Reflexive, Symmetric, and Transitive.
- **Functions:** Every input has exactly one output.
- **Types of Functions:**
 - **One-to-one (Injective):** Distinct inputs have distinct outputs.
 - **Onto (Surjective):** Every element in codomain has a pre-image.
 - **Bijective:** Both one-to-one and onto.
- **Composition of Functions:** $(f \circ g)(x) = f(g(x))$
- **Inverse Function:** If $y = f(x)$, then $x = f^{-1}(y)$. Condition for existence: function must be bijective.

Logarithms

- **Definition:** If $b^y = x$, then $\log_b x = y$.
- **Properties:**
 - $\log_b(xy) = \log_b x + \log_b y$
 - $\log_b(\frac{x}{y}) = \log_b x - \log_b y$
 - $\log_b(x^k) = k \log_b x$
 - $\log_b b = 1, \log_b 1 = 0$
 - **Change of Base:** $\log_b x = \frac{\log_c x}{\log_c b}$

Unit 3: Calculus

Differentiation

- **Definition:** $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$
- **Basic Rules:**
 - $\frac{d}{dx}(c) = 0$
 - $\frac{d}{dx}(x^n) = nx^{n-1}$
 - $\frac{d}{dx}(e^x) = e^x$
 - $\frac{d}{dx}(\ln x) = \frac{1}{x}$
 - $\frac{d}{dx}(cf(x)) = cf'(x)$
 - **Sum/Difference Rule:** $\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$
 - **Product Rule:** $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
 - **Quotient Rule:** $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$
 - **Chain Rule:** $\frac{d}{dx}(f(g(x))) = f'(g(x))g'(x)$

Applications of Derivatives

- **Rate of Change:** If $y = f(x)$, then $\frac{dy}{dx}$ is the rate of change of y with respect to x .
- **Increasing/Decreasing Functions:**
 - $f'(x) > 0 \implies f(x)$ is increasing
 - $f'(x) < 0 \implies f(x)$ is decreasing
- **Local Maxima/Minima (First Derivative Test):**
 - If $f'(c)$ changes from +ve to -ve at c , then c is a local maximum.
 - If $f'(c)$ changes from -ve to +ve at c , then c is a local minimum.
- **Local Maxima/Minima (Second Derivative Test):**
 - If $f'(c) = 0$ and $f''(c) < 0$, then c is a local maximum.
 - If $f'(c) = 0$ and $f''(c) > 0$, then c is a local minimum.
 - If $f'(c) = 0$ and $f''(c) = 0$, test fails (use first derivative test).

Integration

- **Indefinite Integral:** $\int f(x)dx = F(x) + C$ where $F'(x) = f(x)$
- **Basic Rules:**
 - $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ (for $n \neq -1$)
 - $\int \frac{1}{x} dx = \ln|x| + C$
 - $\int e^x dx = e^x + C$
 - $\int cf(x)dx = c \int f(x)dx$
 - $\int (f(x) \pm g(x))dx = \int f(x)dx \pm \int g(x)dx$
- **Definite Integral:** $\int_a^b f(x)dx = F(b) - F(a)$ (Fundamental Theorem of Calculus)

Applications of Integration

- **Area under a curve:** $\int_a^b f(x)dx$ (if $f(x) \geq 0$)
- **Marginal Cost/Revenue:** If $C(x)$ is cost function, $C'(x)$ is marginal cost. $\int C'(x)dx = C(x) + K$.

Unit 4: Probability Distributions

Random Variables

- **Definition:** A variable whose value is a numerical outcome of a random phenomenon.
- **Types:**
 - **Discrete:** Can take on a finite or countably infinite number of values (e.g., number of heads in coin tosses).
 - **Continuous:** Can take on any value within a given range (e.g., height, weight).

Probability Distributions

- **Probability Mass Function (PMF) for Discrete RV:** $P(X = x_i) = p_i$
 - $\sum p_i = 1$
- **Probability Density Function (PDF) for Continuous RV:** $f(x)$
 - $f(x) \geq 0$ for all x
 - $\int_{-\infty}^{\infty} f(x)dx = 1$
 - $P(a \leq X \leq b) = \int_a^b f(x)dx$
- **Cumulative Distribution Function (CDF):** $F(x) = P(X \leq x)$
 - For discrete: $F(x) = \sum_{x_i \leq x} P(X = x_i)$
 - For continuous: $F(x) = \int_{-\infty}^x f(t)dt$

Expected Value and Variance

- **Expected Value (Mean):**
 - Discrete: $E(X) = \sum x_i P(X = x_i)$
 - Continuous: $E(X) = \int_{-\infty}^{\infty} xf(x)dx$
- **Variance:** $Var(X) = E(X^2) - [E(X)]^2$
 - $E(X^2) = \sum x_i^2 P(X = x_i)$ (discrete)
 - $E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx$ (continuous)
- **Standard Deviation:** $\sigma_X = \sqrt{Var(X)}$

Binomial Distribution

- **Conditions:** Fixed number of trials (n), two outcomes (success/failure), independent trials, constant probability of success (p).
- **PMF:** $P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$
- **Mean:** $E(X) = np$
- **Variance:** $Var(X) = np(1-p)$

Poisson Distribution

- **Conditions:** Events occur with a known constant mean rate, independently of time since last event.
- **PMF:** $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$ (where λ is the average number of events in an interval)
- **Mean:** $E(X) = \lambda$
- **Variance:** $Var(X) = \lambda$

Normal Distribution

- **PDF:** $f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$
- **Mean:** μ
- **Variance:** σ^2
- **Standard Normal Distribution:** $Z = \frac{X-\mu}{\sigma}$ (Mean=0, Std Dev=1)
- **Properties:** Symmetric about mean, bell-shaped curve.

Unit 5: Inferential Statistics

Population and Sample

- **Population:** Entire group of interest.
- **Sample:** Subset of the population.
- **Parameter:** Numerical descriptive measure of a population.
- **Statistic:** Numerical descriptive measure of a sample.

Sampling Methods

- **Simple Random Sampling:** Every member has an equal chance of being selected.
- **Systematic Sampling:** Select every k^{th} item after a random start.
- **Stratified Sampling:** Divide population into strata, then simple random sample from each stratum.
- **Cluster Sampling:** Divide population into clusters, randomly select clusters, then sample all members in selected clusters.

Central Limit Theorem (CLT)

- For a large sample size ($n \geq 30$), the sampling distribution of the sample mean ($\bar{X}$) will be approximately normal, regardless of the shape of the population distribution.
- **Mean of sampling distribution of $\bar{X}$:** $E(\bar{X}) = \mu$
- **Standard deviation of sampling distribution of $\bar{X}$ (Standard Error):** $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$

Confidence Intervals

- **Purpose:** To estimate a population parameter using a sample statistic with a certain level of confidence.
- **Formula:** Statistic $\pm$ (Critical Value) $\times$ (Standard Error)
- **For Population Mean (μ) with known σ :**
 - $\bar{X} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$
- **For Population Mean (μ) with unknown σ (use s):**
 - $\bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}$ (use t-distribution for small samples)

Hypothesis Testing

- **Null Hypothesis (H_0):** A statement of no effect or no difference (e.g., $\mu = \mu_0$).
- **Alternative Hypothesis (H_1):** What we want to prove (e.g., $\mu \neq \mu_0, \mu > \mu_0, \mu < \mu_0$).
- **Test Statistic:** Calculated from sample data.
 - **For mean (known σ):** $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$
 - **For mean (unknown σ):** $t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$
- **P-value:** Probability of observing a test statistic as extreme as, or more extreme than, the one calculated, assuming H_0 is true.
- **Decision Rule:**
 - If P-value $\leq \alpha$ (significance level), reject H_0 .
 - If P-value $> \alpha$, fail to reject H_0 .

Unit 6: Time Based Data

Components of Time Series

- **Trend (T):** Long-term increase or decrease.
- **Seasonality (S):** Periodic fluctuations (e.g., daily, weekly, monthly, quarterly).
- **Cyclical (C):** Long-term oscillations not of fixed period.
- **Irregular/Random (I):** Unpredictable variations.

Models

- **Additive Model:** $Y_t = T_t + S_t + C_t + I_t$ (used when absolute magnitude of fluctuations is constant)
- **Multiplicative Model:** $Y_t = T_t \times S_t \times C_t \times I_t$ (used when magnitude of fluctuations varies with level of series)

Moving Averages

- **Simple Moving Average (SMA):** Average of a fixed number of previous data points.
• $MA_t = \frac{Y_{t-k} + \dots + Y_t}{k}$
- **Centered Moving Average:** Used for removing seasonal components. Place the average at the center of the period.

Seasonal Index

- **Calculation:** Actual Value / Seasonal Average.
- **Deseasonalization:** Divide actual values by seasonal indices (for multiplicative model).
- **Forecasting with Seasonal Index:** Forecast (Trend) $\times$ Seasonal Index.

Trend Line Equation

- **Linear Trend:** $Y_t = a + bt$
• $b = \frac{n \sum tY - (\sum t)(\sum Y)}{n \sum t^2 - (\sum t)^2}$
• $a = \bar{Y} - b\bar{t}$

Unit 7: Financial Mathematics

Interest

- **Simple Interest (SI):** $SI = \frac{P \times R \times T}{100}$
• Amount (A) = $P + SI$
- **Compound Interest (CI):** $A = P \left(1 + \frac{R}{100}\right)^n$ (compounded annually)
• If compounded k times a year: $A = P \left(1 + \frac{R}{100k}\right)^{nk}$
• $CI = A - P$
- **Effective Rate of Interest:** Equivalent annual rate when compounding occurs more frequently than once a year.
• $r_{eff} = \left(1 + \frac{R}{k}\right)^k - 1$

Annuities

- **Definition:** A series of equal payments made at regular intervals.
- **Future Value of an Ordinary Annuity:** $FV = P \left[\frac{(1+i)^n - 1}{i} \right]$
• P = periodic payment, i = interest rate per period, n = number of payments.
- **Present Value of an Ordinary Annuity:** $PV = P \left[\frac{1 - (1+i)^{-n}}{i} \right]$
- **Sinking Fund:** A fund into which periodic payments are made to accumulate a specific future sum.
• Periodic Payment $P = FV \left[\frac{i}{(1+i)^n - 1} \right]$

Loans and EMIs

- **Equated Monthly Installment (EMI):**
• $EMI = P \times \frac{i(1+i)^n}{(1+i)^n - 1}$
• P = Principal loan amount, i = monthly interest rate, n = total number of monthly payments.
- **Amortization:** Process of paying off a debt over time in regular installments.

Depreciation

- **Straight-Line Method:** Depreciation per year = $\frac{\text{Original Cost} - \text{Salvage Value}}{\text{Useful Life}}$
- **Written Down Value (WDV) Method:** $D = 1 - \left(\frac{S}{C}\right)^{1/n}$
• D = Rate of Depreciation, S = Salvage Value, C = Original Cost, n = Useful Life.

Unit 8: Linear Programming

Introduction

- **Objective Function:** A linear function to be maximized or minimized (e.g., $Z = ax + by$).
- **Constraints:** Linear inequalities or equalities that limit the feasible region.
- **Decision Variables:** Variables representing the quantities of activities or choices to be made.
- **Feasible Region:** The set of all points that satisfy all constraints. It is a convex polygon.
- **Optimal Solution:** A point in the feasible region that gives the maximum or minimum value of the objective function.

Graphical Method

1. **Formulate the LPP:** Define objective function and constraints.
2. **Graph the Constraints:** Treat inequalities as equalities to plot lines. Shade the feasible region.
3. **Identify Corner Points:** Find the coordinates of the vertices of the feasible region.
4. **Evaluate Objective Function:** Substitute the coordinates of each corner point into the objective function.

5. Determine Optimal Solution:

- For maximization, the corner point giving the largest value of Z is the optimal solution.
- For minimization, the corner point giving the smallest value of Z is the optimal solution.

Special Cases

- **Unbounded Solution:** The feasible region is unbounded, and the objective function can be increased/decreased indefinitely.
- **No Feasible Solution:** The constraints are contradictory, so no common feasible region exists.
- **Multiple Optimal Solutions:** The objective function line is parallel to one of the boundary lines of the feasible region, leading to an infinite number of optimal solutions along that line segment.

@ThunderStudy

